SCST1143 - DATABASE ENGINEERING
SEM 1 – 2024/2025
ASSIGNMENT 1 - EERD

INSTRUCTIONS:
You are required to complete this assignment in a group of **THREE (3)** students.
Each group must submit only **ONE** answer in PDF format via e-Learning.

QUESTION 1

Examine the Employee relation in Table 1, and draw a complete Entity Relationship Diagram (ERD) or Enhanced Entity Relationship Model (EERD) based on the provided information.

Table 1: Employee relation

ENo	*EName*	*BDate*	*Title*	*Salary*	*SupENo*	*DNo*	*Lang*	*DB*	*MBA*	*Bonus*
E1	Dolly	01-05-75	EE	15000	E2	D3	C++	MySql	Y	4000
E2	Sarah	06-04-79	SA	13000	E5	D3			N	2000
E3	Leman	07-05-76	ME	10000	E7	D2			N	3000
E4	Mika	18-10-71	PR	15000	E6	D3	Java	Oracle	Y	4000
E5	Choon	06-04-79	SA	12000	E8	D3			Y	4000
E6	Cun	30-09-73	EE	14000	E7	D2	C++	DB2	Y	3500
E7	David	09-08-77	ME	12000	E8	D1			N	3000
E8	Jaffar	10-11-72	SA	16000		D1			Y	6000

QUESTION 2

ImpianJasa Sdn. Bhd. is a company that processes applications for potential employees with appropriate skills. Based on the following descriptions, draw an Enhanced Entity Relationship Diagram (EERD):

a) Applications for potential employees received by ImpianJasa Sdn. Bhd. are categorized as IT-based companies or non-IT-based companies.
b) The skills required by IT-based companies for potential employees include technical and generic skills.
c) For non-IT-based companies, potential employees are not necessarily required to have programming or PC repair skills.

QUESTION 3

Al-Zahra Training Centre is a company that conducts training courses. As a junior data admin at Maju Teguh Software House, you are required to design a conceptual database design. The following are the business rules for which you need to create a complete Enhanced Entity Relationship Diagram (EERD):

- For each trainee (approximately 2,000 trainees), identified by a unique trainee code, we need to store their name, age, sex, and place of birth.
- We also need to store information about each trainee's previous employers, including the employer's name, address, and telephone number, as well as the trainee's employment periods (determined by the start and end dates of past employment).
- To work with Al-Zahra Training Centre, each trainee must have at least one employment history record as a prerequisite.
- Some trainees may have worked at the same companies in the past, particularly at companies like Oracle and Microsoft.
- Trainees are classified as either **self-employed** or **employees**.
    - If a trainee is **self-employed** (e.g., a professional), we need to record their area of expertise and, if applicable, their professional title.
    - If a trainee is an **employee**, we need to store their job level and the position they held in the company.

Based on the above business rules, draw a complete Enhanced Entity Relationship Diagram (EERD) for Al-Zahra Training Centre.

QUESTION 4

Draw an Enhanced Entity Relationship Diagram (EERD) based on the following descriptions:

- Some of the permanent staff are categorized as Admin, R&D or Finance personnel.
- A permanent staff member can belong to both Admin and Finance departments simultaneosly.
- A staff member can either be a manager or be supervised, but not both.

QUESTION 5

Draw an Enhanced Entity Relationship Diagram (EERD) based on the following descriptions:

> "A publication management system manages different types of publications. The majority of these publications are in the forms of journal articles, textbooks and novels. All journal articles are categorized into indexed and non-indexed publications."

QUESTION 6

Consifer the following requirements for the ABC Sdn. Bhd. database, which is used to manage staff information:

- ABC Sdn. Bhd. tracks of each staff member's number, name (first and last name), current address and phone number.
- Some applications also require the city, state and ZIP code associated with the staff's current address.
- Data stored about vehicles includes the vehicle number, model and price.
- Each staff member owns one or more vehicles, which are categorized as either a car or a motocycle.

Based on the requirements above, draw a conceptual Entity Relationship Diagram (ERD) or Enhanced Entity Relationship Diagram (EERD).